LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY

(UGC Autonomous)

Approved by AICTE | Affiliated to Osmania University | Estd.2003

Accredited by **NBA**, Accredited '**A**' Grade by NAAC

Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: VII **Section:** A

Project Group ID: CSDMP230008

THE ECOLOGICAL AND FINANCIAL IMPACT STUDY OF ELECTRIC BUS TRANSITION

Abstract:

The transition from conventional diesel buses to electric buses represents an important approach toward achieving cleaner, more efficient, and economically sustainable public transportation. This project focuses on evaluating the ecological and financial impacts of adopting electric buses through an integrated analytical framework. The proposed system combines greenhouse gas (GHG) emission analysis, real-time data analytics, and discounted cash flow (DCF) analysis to provide a comprehensive assessment of the environmental and economic feasibility of electric bus transition. CO2-equivalent emissions are calculated using parameters such as fuel consumption, vehicle efficiency, and operational conditions, enabling comparative evaluation of diesel and electric buses. A centralized digital platform facilitates data collection, processing, visualization, monitoring, and report generation, thereby improving transparency, scalability, and decision-making. The system also enables stakeholders to evaluate environmental benefits and financial outcomes before implementing large-scale electric bus fleets.The proposed project contributes to **SDG 7:** Affordable and Clean Energy by promoting the adoption of clean-energy-based transportation. It supports **SDG 11:** Sustainable Cities and Communities by encouraging sustainable and environmentally responsible public mobility, while contributing to **SDG 13:** Climate Action through the reduction and monitoring of greenhouse gas emissions.

Keywords: Electric Buses, GHG Emissions, Financial Analysis, Sustainable Transportation, ML

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor:

Signature: _____ **Date**: _____

Coordinator **Head of the Department**